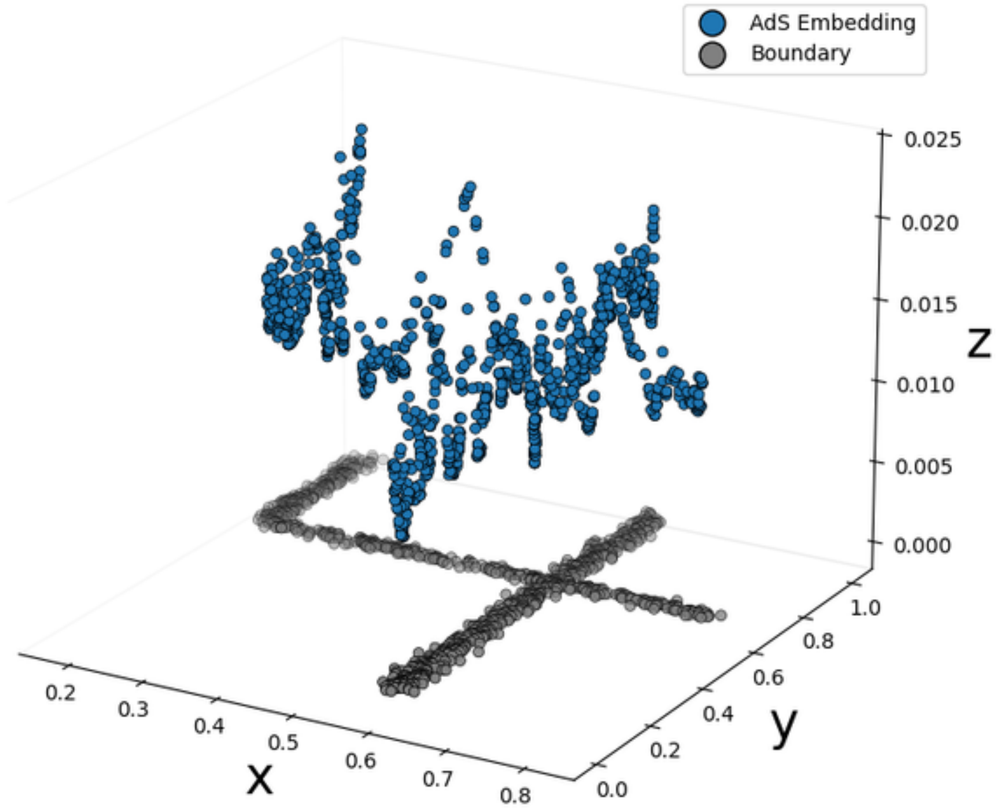
AdS-GNN: Lifting

$$ds^2 = \frac{1}{z^2}(dz^2 + dx^2 + dy^2)$$



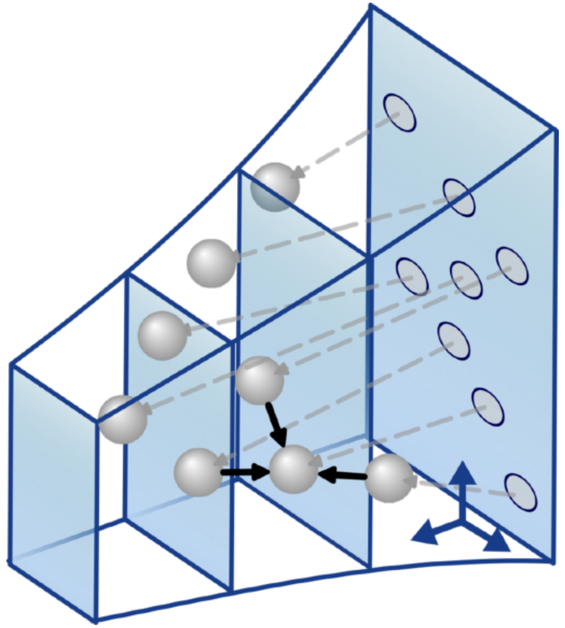
Need to assign a z_i value to each point.

This can be interpreted as a **length scale**: obtain this by averaging distance from its *k* nearest neighbours in a covariant way.

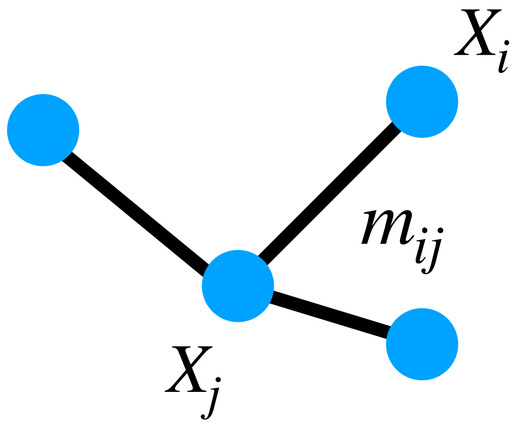
Points in low-density regions end up further in the interior.

AdS-GNN: Bulk operations

Now we have a graph of points living in AdS space.



Act on it with **message-passing**; points talk to their neighbours in a way that depends only on intrinsic AdS proper distance.



Final information aggregated over AdS or sent back to boundary: **conformally equivariant neural network**.